

Reflective Lantern - Innovation Report

New Project Built

Name: Suite-Cast

Branch: <https://github.com/atharvadevne123/reflective-lantern/tree/innovation/suite-cast>

Pull Request: <https://github.com/atharvadevne123/reflective-lantern/pull/22>

Date: 2026-07-15

Inspired by: Hospitality revenue-management gap in portfolio (HN/GitHub trending blocked by egress policy; topic chosen via duplicate-gate elimination)

What it does: Hotel booking demand forecasting and dynamic pricing API using an XGBoost-LightGBM ensemble with 18 engineered features, KS-test drift monitoring, SQLAlchemy prediction logging, and champion/challenger retraining.

Note: session GitHub access is scoped to reflective-lantern, so standalone repo creation was blocked. Project shipped on branch innovation/suite-cast with draft PR #22 (same approach as Energy-Seer and Traffic-Pulse). The release tag suite-cast-v1.0.0 exists locally but tag pushes are blocked by the git proxy.

Tech Stack Coverage: 100%

- [X] Python
- [X] FastAPI/Flask
- [X] XGBoost/LightGBM/RF
- [X] sklearn Pipeline
- [X] Model monitoring
- [X] Docker+compose
- [X] SQLAlchemy
- [X] pytest suite
- [X] .env.example
- [X] GitHub Actions CI

Optional picks (4+): time-series/seasonality features, drift detection, automated retraining (Airflow champion/challenger DAG), 5-fold CV + AUC-ROC, Alembic migrations, rate limiting, correlation-ID tracing.

Project Stats

Files created: 37

Tests written: 100 (all passing)

Commits: 60

Lint: ruff check + format clean